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Noise Generated by Raindrop on Metal Deck Roof Profiles: It's Effect towards People Activities

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Abstract

Noise generated by rain fall on un-insulated metal deck roofs can cause serious acoustic problems in buildings. This paper reports the outcomes of an effect of sound created by rain fall on un-insulated metal deck roofs towards people's activities. The measurement and questionnaire distribution was done at selected UiTM buildings that have been built using un-insulated metal deck roofs. It was found that the occupants facing inconveniences condition that can disturb their activities. The comfort noise level that accepted by the human being is between 55 to 60dB. Therefore, this study substantiates the notion that rain noise can contribute to negative effects in people's lives during rain.

*Keywords:*Rainfall intensity; raindrop noise level; metal roofing profile and speech interference level

1. Introduction

Sound generated by rain fall can become unpleasant to the surrounding area depending on the surfaces on which it falls. It becomes more disturbed when it falls on metal deck roof profiles. It has been noted that noise level that can be accepted by human hearing is between 55dB to 60 dB (Bronzaft and Hagler,

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2010). If it goes higher than that, it is likely to create annoying noise affecting daily human activities that can cause an uncomfortable condition towards their lives. Building Bulletin 93, London (2004) reports the level of rain noise on metal deck can exceed to 70 dB.

According to World Health Organization (1995) quality of life defines as “individual’s perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns." It refers to a combination of positive characteristics that influence by physical, psychology and social nature. Usually, these characteristics may lead to people life more well-being (Irene et al, 2003) and valuable.

However, this situation becomes vice versa when the other elements such as noisy environment take place. Michael & Robert (2010) discovered noise exposure may lead to adverse health effects because it often disturbs basic people activities such as concentration, ability to perform, rest, sleep and communication (Philomena, 2010). Probably, it leads to annoyance feeling that triggers stress, unfocused, communication failure and disrupts to sleep that significantly reduces the value of quality of people’s lives.

Universiti Teknologi Mara (UiTM) is one of the institutions that used metal deck roof profiles as a main roof system in their buildings. Almost 90 percent of the total buildings in UiTM Campuses have been built using metal deck roof profiles. It is likely that during rain, the noise created by these roofs may cause inconveniences to the occupant of these buildings. Hence, the objectives of this paper are to measure the rain noise level transmitted through un-insulated metal deck roof profiles and to evaluate its effects towards people activities.

2. Previous Research Findings on the Rain Fall Noise

A striking rapid raindrop on metal deck roofs significantly increases ambient noise, reverberation time and reduces speech intelligibility in a building. Investigations have done by Carter et al (2002); Building Bulletin 93, London (2004); Andy et al (2007) showed that buildings that used this system faced serious acoustic problems. Besides that, the level of noise produced by raindrop impact on metal deck roofs can exceed 70 dB (Mariam & Faris, 2005; Building Bulletin, 93 2004; Ballagh, 1990) and this level is higher compared to the areas exposed to aircraft noise (Daniel et. al. 2010) or traffic noise (Katarina et al, 2009) which is only up-to 65dB. Thus, the noise range produced by rain fall on metal deck roof systems can cause discomfort to people.

From a comprehensive investigation done by Lee, (2004), it is discovered that the impact from rain noise can be categorized as the irritating noise causing disturbance and annoyance. Often theaters, stadiums, auditoriums, classrooms, educational institutions, sport complexes and commercial buildings faced such problems because they are often built with such roof claddings due to the need for larger spans. Dubout, (1969); Ballagh (1990); Andy et al, (2007) on rain noise have pointed out that rain contributes to noise and vibration to a structure or a building. This noise and vibration are transmitted through the roof and is radiated as structural air-borne noise into a building. These radiations may significantly increase ambient noise in a building (Carter et al, 2002).

Furthermore, Andy (2007) pointed out that the impact from the rain fall onto a roof can reduce the speech intelligibility among occupants in a building. If this happens in-school buildings, it might impede children’s ability to hear and therefore, understanding in a classroom (Shield and Dockrell, 2003) and affect children’s learning and development (Marie et al, 2002).

However, these issues have not received much attention until recently in Malaysia and not much has been done to investigate the noise issues related to the rain noise transmitted on the metal deck roofs. In fact, Malaysia is one of the countries that have a high annual rainfall, and it shows that noise levels caused by rapidly raindrops on light weight metal roofs are not acceptable. So this paper attempt to find out the range of noise that generated by raindrop impact on metal deck roof profiles without insulation and its effect to peoples' activities.

3. Rain Intensity

Because of these study involves a rain parameters, understanding of rain intensity is important. To classify the rain intensity i.e. mm, Table 1 of MS IEC 60721-2-2 2004 on the classification of rainfall was referred. The extract of Table 1 is as below;

Table 1. Classification of rain intensity according to MS IEC 60121-2-2. Sources: MS IEC 60121-2-2 2004 Classification of Environmental Conditions Part 2: Environmental Condition Appearing in Natural Precipitation and Wind

Rainfall Type	Rainfall Rate mm
Moderate	Up to 4
Intense	Up to 15
Heavy	Up to 40
Cloudburst	Greater than 100

4. Brief survey on selected buildings.

A survey on UiTM buildings was conducted in order to select buildings that use the same types and designs of un-insulated metal deck roof profiles. Three buildings were selected, and it was found that all the buildings were covered with metal deck roof system and no ceiling are fixed underneath. Two of the buildings were built with flat and curved roofs which are not insulated. Similarly, another building with a pitch roof was also not insulated.

5. Brief methodology

This research employed two methodologies, they are:

- Real-Time Measurement on the noise level created by the raindrops.
- Questionnaire distributed to 60 respondents to determine their responses on the noise created by the raindrops.

For the first task, three buildings in the UiTM Shah Alam Campus were selected with metal deck roof systems, neither thermal nor sound insulated and with no ceilings fixed underneath. The buildings are:

- Swimming Pool Complex UiTM - Flat and curve roof
- Mutiara Bistari Kindergarten, UiTM – Pitch roof
- Sports complex, UiTM – Flat and pitch roof

Open space underneath the roof of each of the buildings was chosen where the measurement of the transmitted rain noise through the roof systems were recorded using Sound Level Meter (SLM). The duration of each noise measurement was 40 minutes, and rain intensity was 60 minutes. During the measurement, the SLM was mounted on a tripod, 1.2 m high from the floor, in the centre of space selected in the buildings and away from any reflective surfaces. At the same time, the intensity of rain was measured using a traditional rain gauge and a measuring cylinder. The rain gauge was placed on a tripod about 500mm from the ground level.

60 questionnaires were distributed to the users of the buildings studied. The questionnaire consists of three major parts. Part one is on respondents demographic. Part 2 consists of the awareness on the rain and rain noise, and Part 3 is regarding to affect of rain noise towards peoples' activities. All the questionnaires were distributed and collected by hand. The responses from the questionnaires were analysed using SPSS software.

6. Measurements of the Rain Intensity and its Noise Level

The days and times when measurements were conducted at the buildings under-studied were raining time. Time of noise level measurement was 40 minutes and raining duration 60 minutes. The rain noise levels recorded at single octave frequency range from 125 Hz to 8 kHz. The rain intensities and noise level recorded were shown below (Table 2).

Table 2. Rain intensities (mm) and Noise Level at (1/1) Single Octave Frequency Range

Type of Building	Rain intensity (mm)	Frequency						
		125	250	500	1k	2k	4k	8k
Swimming pool Complex	(8) Intense	68.2	65.8	67.7	64.5	61.9	58.4	51.9
Mutiara Bistari Kindergarten	(20) Heavy	84.4	84	81.9	81.8	79.7	74.6	69.4
Sport Complex UiTM	(27) Heavy	75.5	76	79.2	82.9	81.8	79.8	72.4

The average noise level at frequency range of 500Hz, 1 kHz and 2 kHz for Swimming Pool Complex, Mutiara Bistari Kindergarten and Sport Complex building is 64.7dB, 81.1 dB and 81.3dB respectively.

Table 3. Amplitude distribution

Type of Building	Amplitude distribution (%)								
	50.0-55.0	55.0-60.0	60.0-65.0	65.0-70.0	70.0-75.0	75.0-80.0	80.0-85.0	85.0-90.0	90.0-95.0
Swimming pool Complex	0	0	0	75	25	0	0	0	0
Mutiara Bistari Kindergarten	0	0	0	0	12.5	12.5	12.5	62.5	0
Sport Complex UiTM	0	0	0	0	0	0	37.5	50.0	12.5

The noise amplitude distributions for all types of buildings are shown in Table 3 which shows that 100 percent of the noise levels exceed 55dB.

7. Analysis of the Questionnaires

7.1. Part A: Demographic of the respondents

From the analysis of the survey, 80 percent of the respondents were between the ages of 20 to 30 years old, 27 of them are male and 33 female and 66.7 percent or 40 of them are students and 20 staff. 60 sets of questionnaire were distributed, all of them were returned and analysed.

7.2. Part B: Awareness on the rain and rain noise

98 percent of the respondents were aware that it was raining and 50 percent of them agreed that the noise created by the rain was disturbing. Almost 80 percent of the respondents classified sound created by the rain as noise. But almost all of them did not realize that the noise created by the rain drops increased because of the types of roofs that the buildings have.

7.3. Part C: Effects of rain noise towards people's activities

When asked what are the main causes that will hinder their work or study when it rains, 52 percent of the respondents agreed that one of the main causes is noise from the roof and 48 percent of them agreed that rain will hinder their movement. On top of that, 75 percent of the respondents agreed that the rain noise affects their work or study. To find out the serious effects of the rain noise, the responses are summarized in Table 4 However, 25 percent of the respondents think that noise from the rain does not affect their activities.

Table 4. Serious Effect of Rain Noise to people

Survey	Effect of rain noise to people	Percentage of Response (%)
Serious effetcs of rain noise	Unfoccused	43
	Annoyance	30
	Stress	2
	Not affect their activities	25
Communication problem	Hamper communication	82
Is it necessary to elevate your voice	(Yes) To elevate voice	82

8. Discussion

The Recorded rain noise levels, shows that the level of noise generated by rain fall on un-insulated metal deck roofs of the buildings studied increases the background noise in those buildings extremely. Table 5 summarizes the results corresponding to the rain intensity measured. As noted earlier, background noise levels that can be accepted by human hearing is between 55dB to 60 dB (Bronzaft and Hagler, 2010) and all the measurement results obtained all exceeded the limit.

Table 5. The maximum and minimum L_{eq} of rain noise against various rain intensities

Buildings	Rain fall rate (mm)	Maximum level of noise, Leq (dB)	Minimum level of noise, Leq (dB)
Swimming pool complex	8 (intense)	64.7	66.3
Mutiara Bistari Kindergarten	27 (heavy)	89.9	72.4
Sport complex	20 (heavy)	91.2	82.4

Background noise will inevitably have a masking effect on speech. The higher the background noise level, the more the voice will have to be raised for satisfactory speech communication. Speech interference levels (SIL) are a quantitative assessment of these parameters, expressing the masking effect of the background noise to the conversation between people. SIL is the arithmetic average of sound pressure levels of background noise in the octave band center frequencies of 500, 1000 and 2000Hz. Once the SIL is determined, it can be used to assess the ideal distance from the speaker to the listener during conversation. Table 6, proposed by K.S.Pearson, R.L.Bennet and S.Fidell (1977) below can be used as guidelines.

Table 6. Speech Interference Levels. Source: Pearson. K.S., Bennet. R,L, & Fidell. S, 1977

Distance	Background Noise Levels (dB)					
(m)	Whisper	Low	Normal	Raised	Very loud	Shouting
0.25	41	56	67	73	79	85
0.50	36	50	61	67	73	79
0.75	32	47	57	63	69	75
1.00	29	44	55	61	67	73
1.50	26	41	51	57	63	69
2	-	38	49	56	61	67
3	-	-	45	51	57	63
4	-	-	43	49	55	61
5	-	-	41	47	53	59
6	-	-	39	45	51	57

Calculated SILs for the spaces of the buildings studied are 72.4, 81.1 and 81.3 dB respectively as shown in Table 7 below. If the speaker and listener are having a conversation at a distance of 1m in these background noise levels, and by referring to Table 7, they have to speak either very loud or even shout to each other.

Table 7. Calculated Speech Interference Levels

Buildings	SIL (dB)	Distance between the Speakers (m)	Level of Voice
Swimming pool complex	72.4	1	Very loud
Mutiara Bistari Kindergarten	81.1	1	Shouting
Sport complex	81.3	1	Shouting

This is supported by the results of the questionnaires which show 82 percent of the respondents agreed that raining noise can hamper their communication and they have to elevate or raise their voice if they would like to continue talking during raining days. The results of the questionnaires also show that during heavy rains, 43 percent of them loose focus to what they do, 30 percent found that the rain noise is very annoying and only 2 percent thought that rains may cause stress. It shows that, this exposure may give negative effect towards peoples' activities such as communication failure; lose concentration and reduces ability to perform.

9. Conclusion

Since the popularity of metal deck roof profiles was increasing in current building practices, the exposure of noise generated by rapid rain drop has extremely increased. The real rain noise level measurements show that most of the buildings that use metal deck roof profiles generated a high level of noise in the spaces underneath when it rains.

The findings from the respondents of the questionnaires indicate that they were aware about these issues and classified sound from rain as noise that produces negatives effects on their work or study. Sound generated by rain drops also produces serious effects to the respondents psychologically. Most of the respondents became unfocused and annoyance effects during raining time. The findings also discovered that rain noise may affect conversation or communication among the users of a building. Almost 82 percent of them agreed that they also need to speak loud or even shouting if they were to continue talking.

Therefore, the result shows that sound generated by rain drop on metal deck roof profiles contribute serious acoustic problems to a building and produce a negative effect in people lives during rain, in terms of psychological and communication problems.

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